

Question Paper

Exam Date & Time: 22-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

COMPUTER ORGANIZATION and ARCHITECTURE [CSE_2121]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) In a robotics laboratory, a CISC-based processor is programmed to compute the dot product of two (4)
sensor data arrays, SENSOR_A and SENSOR_B, each containing N elements.
A) The processor loads the base addresses of both arrays and initializes a counter COUNT with the
value N. During each iteration of the loop, it performs the following operations:
1. Loads one element from each array.
 2. Multiplies the two elements.
 3. Adds the product to an accumulator register that stores the running total.
 4. Decrements the counter COUNT.
 5. Repeats the process until all pairs of elements are processed.

After the loop completes, the final accumulated value is stored in the memory location
DOT_PRODUCT.

Describe the sequence of operations needed to perform this task using appropriate assembly-style
instructions and provide suitable explanations for each step.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Determine the product of the following numbers by employing the Booth's multiplication algorithm: (4)
M = -9 and Q = 15

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Convert each of the following pairs of decimal numbers into 6-bit 2's complement representation (2)
and perform the specified arithmetic operation. For each case, determine whether overflow occurs.
If overflow is present, explain the reason for its occurrence.
i) Add 4 and 11
ii) Subtract -9 from -2

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2) Evaluate the effectiveness of Chopping and Rounding methods in minimizing truncation errors in (4)
floating point representations, and justify which method provides higher numerical accuracy with
suitable examples.
A)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- B) In context of the hardwired control unit design, draw the state diagram of the Booth's Multiplier (4)
Controller along with the steps in its controller action. In what ways, the Hardwired Control Design
Mechanism is disadvantageous in comparison with that of the Microprogrammed Control Unit
Design Mechanisms.

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

- C) Depict the Hardware Implementation of Multiplication of unsigned Binary Numbers. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 3) A computer has a small data cache capable of holding eight words and has a byte addressable memory. Each cache block consists of one word and one word comprises of one byte. When a given program is executed, the processor reads data sequentially from the addresses given below (Addresses are given in decimal). Assume that the cache is initially empty. Show the contents of the cache after each memory access, for a Direct mapped cache, and compute the hit rate 7200, 7201, 7208, 7213, 7200, 7216, 7208, 7201, 7215, 7201, 7213, 7218, 7220, 7213, 7217, 7219 (4)
- A)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- B) Demonstrate how virtual memory and address translation operate in a computer system by explaining the step-by-step process of translating a virtual address to a physical address using an example memory mapping scenario. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Draw neat diagrams of one bus and three bus architectures. Analyse the number of clock cycles to $R2 \leftarrow R0 + R1$ assuming that the source and destination are the general purpose registers (2)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4) Explain the concept how DMA data transfer that happens between various I/O devices and memory. Also elaborate on the role of processor, DMA controller in this transfer operation with the help of a suitable diagram. (4)
- A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Analyse the process of DMA data transfer between I/O devices and memory, explaining each step in the operation. Examine the specific roles played by the processor and the DMA controller during this transfer and illustrate the process of DMA controllers with an appropriate diagram. (4)

Course Outcome: "CO4" / Cognitive Level: "Evaluate"

- C) Distinguish between UMA and NUMA multiprocessor as far as network latency is concerned. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) Specify the need of pipelining in modern day processors. List the advantages of using pipelining. Also consider a five-stage pipelined processor (with stages: Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Write Back (WB)), assume that each stage takes 1 cycle to complete. An instruction sequence consists of five independent instructions. How many cycles will it take for the processor to complete all five instructions? Justify your answer with a proper diagram and explanation. (4)
- A)

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) Analyze the differences between Memory-Mapped I/O and I/O-Mapped I/O addressing techniques in terms of their addressing method, instruction usage, and impact on system performance. (4)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- C) Consider a multiprocessor system using a cache coherence protocol. If Processor P1 writes to a cache line, and Processor P2 has the same cache line in its cache in the Shared state, what happens to Processor P2's cache line when P1 performs the write? Explain the scenario. (2)

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Event Name	Subject
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058-END SEMSTER-ODD-2025 CSE_2121:COMPUTER ORGANIZATION and
ARCHITECTURE

#	Course Outcome	Description	Marks
1	1	CO 1	14
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	12
5	5	CO 5	4
Total			50

* **NA** - CO Not Selected

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058-END SEMSTER-ODD-2025 CSE_2121:COMPUTER ORGANIZATION and
ARCHITECTURE

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	10(20%)
3	Application	3	12(24%)
4	Analysis	3	12(24%)
5	Create	4	16(32%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)